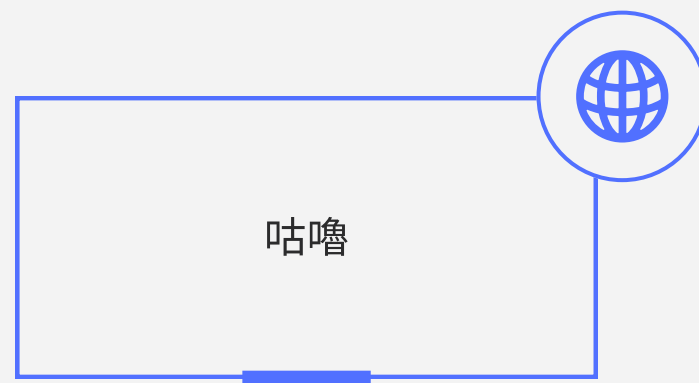
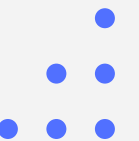
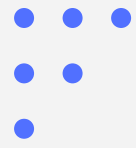


115-1 python 與 微控制器

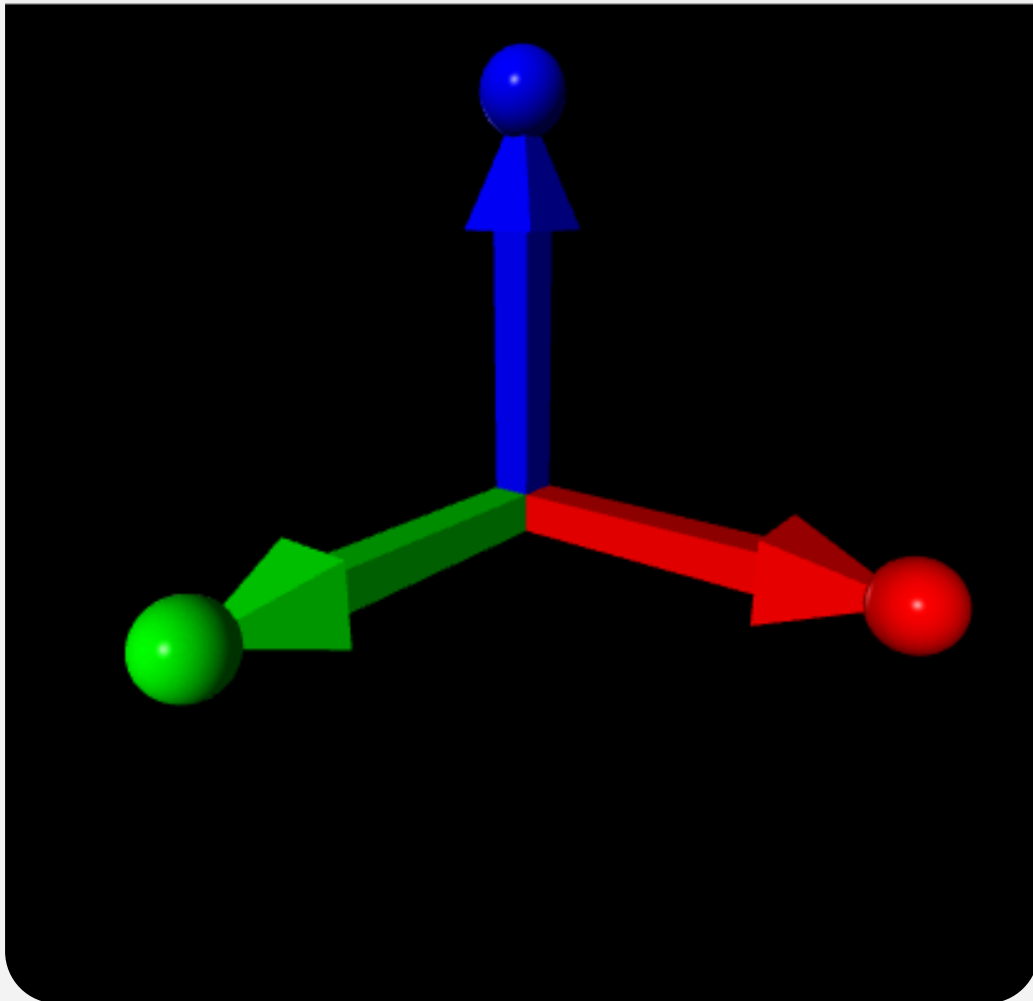


Presented by
高一丙18陶以恩





2026/09/11 09:10:29
ram Screenshot



```
1 Web VPython 3.2
2 |
3 a1 = arrow(pos=vec(0, 0, 0), axis=vec(1, 0, 0), color=color.red)
4 a2 = arrow(pos=vec(0, 0, 0), axis=vec(0, 1, 0), color=color.blue)
5 a3 = arrow(pos=vec(0, 0, 0), axis=vec(0, 0, 1), color=color.green)
6 b1 = sphere(pos=vec(1, 0, 0), radius=0.1, color=color.red)
7 b2 = sphere(pos=vec(0, 1, 0), radius=0.1, color=color.blue)
8 b3 = sphere(pos=vec(0, 0, 1), radius=0.1, color=color.green)
```

20260911-在
vpython 建立附件
箭头+球物件

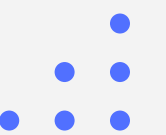


python

main.py

```
1 num = 5
2
3 while num >= 1:
4     print(f"Count: {num}")
5     num -= 1
6
7 print("發射！")
```

建立一個變數 `num = 5`，使用 `while` 迴圈讓程式從 5 倒數到 1（印出 `Count: 5`、`Count: 4` ...），迴圈結束後在最外層印出 "發射！"。





```
1 Web VPython 3.2
2
3 S1 = sphere(pos=vec(0, 0, 0), radius=2, color=color.cyan,opacity=
4
5 count = 0
6 while (count < 5):
7
8     count = count + 1
9
10
11
12 while (True):
13     rate(60)
14     S1.pos.y =S1.pos.y + 0.1
15     S1.radius =S1.radius + 0.01
16     S1.opacity =S1.opacity - 0.01
17
```

— 20260918- 讓一個
球球 向右動 然後消
失 --> 向 "右上" 動



python

```
1 Web VPython 3.2
2
3 S1 = sphere(pos=vec(-10, 10, 0), radius=2, color=color.cyan, opacity=1)
4
5 dt = 0.01
6 g = -10
7
8 vx = 5
9 vy = 0
10
11
12 while S1.pos.y > -10:
13     rate(100)
14
15     S1.pos.x = S1.pos.x + vx * dt
16     S1.pos.y = S1.pos.y + vy * dt
17
18     vy = vy + g * dt
```

20260918-鉛直上拋 改平拋





```
1 Web VPython 3.2
2
3 ground = box(pos=vec(0, -10, 0), size=vec(60, 0.5, 10), color=color.green)
4 S1 = sphere(pos=vec(-20, -8, 0), radius=1, color=color.cyan, make_trail=True, trail_type="curve")
5
6 dt = 0.01
7 g = -10
8
9 vx = 10
10 vy = 15
11
12 while S1.pos.y >= (ground.pos.y + S1.radius):
13     rate(100)
14
15     S1.pos.x = S1.pos.x + vx * dt
16     S1.pos.y = S1.pos.y + vy * dt
17
18     vy = vy + g * dt
```

— 20260918- 平抛
改斜抛